Arithmetic fundamental lemmas and non-reductive geometry

Zhiyu Zhang

Stanford University

07/09/2024, workshop on Branching Problems for Representations of Real, P-Adic and Adelic Groups, BIRS-UBCO Kelowna, Canada

Application of representation theory to arithmetic geometry

In number theory, we study arithmetic invariants of polynomial equations e.g. conductors, counting solutions, L-functions:

$$\{\textbf{polynomial equations (motives) over } \mathbb{Q}\} \to \{\textbf{invariants}\}.$$

For any reductive group G over $\mathbb{Q}$, we expect a Langlands correspondence

$$\mathcal{A}(G) \to \mathcal{B}(G^\vee),\ \pi \mapsto \rho_\pi$$

$$\{\textbf{automorphic representations of } G\} \to \{G^\vee\textbf{-motives over } \mathbb{Q}\}.$$

Application of representation theory to arithmetic geometry

In number theory, we study arithmetic invariants of polynomial equations e.g. conductors, counting solutions, L-functions:

$$\{\textbf{polynomial equations (motives) over } \mathbb{Q}\} \to \{\textbf{invariants}\}.$$

For any reductive group G over $\mathbb{Q}$, we expect a Langlands correspondence

$$\mathcal{A}(G) \to \mathcal{B}(G^\vee),\ \pi \mapsto \rho_\pi$$

$$\{\textbf{automorphic representations of } G\} \to \{G^\vee\textbf{-motives over } \mathbb{Q}\}.$$

Hope

- *(Langlands) Good polynomial equations are automorphic.*
- *(Relative Langlands) Good invariants are automorphic.*

Applications (using Shimura varieties): Fermat's last theorem ($G = \mathrm{GL}_2$), BSD conjecture via Gross-Zagier formula + Euler systems ($G = \mathrm{GSp}_4$, $G = \mathrm{GL}_{2n}$, $\mathrm{GL}_n \times \mathrm{GL}_m$ [GGP]), Colmez conjecture ($G = \mathrm{U}(n)$).

Invariants from symmetry: periods and L-functions

From a good G-variety X e.g. $X = G/H$ for a subgroup $H \leq G$, we get an automorphic invariant called **period integral** for $\pi \in \mathcal{A}_{cusp}(G)$:

$$P_X(\varphi) := \int_{H(\mathbb{A})/H(\mathbb{Q})} \varphi(h)dh, \varphi \in \pi.$$

From a good $G^\vee$-variety $X^\vee$ e.g. a representation of $G^\vee$, we get an arithmetic invariant called **L-functions** for $\rho_\pi \in \mathcal{B}(G^\vee)$:

$$L_{X^\vee}(\rho_\pi, s) := L_{X^\vee,\infty}(\rho_\pi, s) \prod_{p<\infty} \det(1 - p^{-s}\rho_\pi(\mathrm{Frob}_p)|_{X^\vee})^{-1}, s \in \mathbb{C}.$$

Relative Langlands program

Conjecture (Relative Langlands duality, after BZSV)

- *The dual $G \to G^\vee$ extends to good $X \to X^\vee$.*
- *(**Global GGP type conjecture**) Central L-values $L_{X^\vee}(\rho_\pi, 1/2)$ matches (the norm of) $P_X(\varphi)$ up to local factors depending on φ.*

How to prove such conjectures? A powerful way is **relative trace formulas and fundamental lemmas**, see e.g. the recent proof of full unitary GGP conjecture.

Relative Langlands program

Conjecture (Relative Langlands duality, after BZSV)

- *The dual $G \to G^\vee$ extends to good $X \to X^\vee$.*
- *(**Global GGP type conjecture**) Central L-values $L_{X^\vee}(\rho_\pi, 1/2)$ matches (the norm of) $P_X(\varphi)$ up to local factors depending on φ.*

How to prove such conjectures? A powerful way is **relative trace formulas and fundamental lemmas**, see e.g. the recent proof of full unitary GGP conjecture.

- Instead of working on a single π, study relative trace distributions $\mathrm{RTF}(f)$ on whole spectrum $f \in C_c^\infty(G(\mathbb{A}))$.
- Give a spectral decomposition of $\mathrm{RTF}(f)$ into period integrals.
- Give a geometric decomposition of $\mathrm{RTF}(f)$ into sums of global orbital integrals $=$ products of local orbital integrals (nice functions on $G(\mathbb{Q}_p)$), with regular terms $+$ singular terms.
- Establish relations of different local orbital integrals via local analysis: fundamental lemmas and smooth transfers.

Branching laws and signs of L-functions

For $X = G/H$, period integrals $P_X|_\pi \in \mathrm{Hom}_H(\pi, 1)$. In particular,

$$P_X|_\pi \neq 0 \implies \mathrm{Hom}_{H_p}(\pi_p, 1) \neq 0, \forall p \leq \infty.$$

For a $G^\vee$-representation $X^\vee$, the L-function $L_{X^\vee}(\rho_\pi, s)$ is expected to have a functional equation with symmetry $s \to 1 - s$ with sign (epsilon factor) $\epsilon_{X^\vee}(\pi) = \prod_p \epsilon_{X^\vee,p}(\pi) \in \{\pm 1\}$.

Local GGP type conjecture:

$$\mathrm{Hom}_{H_p}(\pi_p, 1) \neq 0 \implies \epsilon_{X^\vee,p}(\pi) = 1, \forall p \leq \infty.$$

Then $P_X|_\pi \neq 0 \implies \epsilon_{X^\vee}(\pi) = 1$, compatible with $L_{X^\vee}(\rho_\pi, 1/2) \neq 0$.

Today

Question: how about polynomial equations with sign $\epsilon_{X^\vee}(\pi) = -1$? In general, how do you understand higher derivatives (coefficients in the Taylor expansion) of L-functions?

Idea from Gross-Zagier formula: use arithmetic analogs of period integrals via X-special cycles on locally symmetric spaces.

Today

Question: how about polynomial equations with sign $\epsilon_{X^\vee}(\pi) = -1$?
In general, how do you understand higher derivatives (coefficients in the Taylor expansion) of L-functions?
Idea from Gross-Zagier formula: use arithmetic analogs of period integrals via X-special cycles on locally symmetric spaces.

- Introduce **local X-special cycles when X is a nice representation of G**, which will play a role in arithmetic relative Langlands program. They are related to mirabolic subgroups of GL_n.
- As an example, explain the **twisted arithmetic fundamental lemma** in the context of the twisted GGP conjecture.

Producing arithmetic invariants from big G-representations

- Step 1: find a nice $G(\mathbb{A}_f)$-representation V_X. Assume that our representation π appears in V_X.
- Step 2: find a good linear functional $\ell_X \in \mathrm{Hom}(V_X, 1)$, which is related to branching laws e.g. $\ell_X \in \mathrm{Hom}_H(V_X, 1)$ for $X = G/H$. Obtain an invariant on $\varphi_f \in \pi_f$ as a restriction of ℓ_X to π.

Producing arithmetic invariants from big G-representations

- Step 1: find a nice $G(\mathbb{A}_f)$-representation V_X. Assume that our representation π appears in V_X.
- Step 2: find a good linear functional $\ell_X \in \mathrm{Hom}(V_X, 1)$, which is related to branching laws e.g. $\ell_X \in \mathrm{Hom}_H(V_X, 1)$ for $X = G/H$. Obtain an invariant on $\varphi_f \in \pi_f$ as a restriction of ℓ_X to π.

Period integrals for $X = G/H$: $V_X = L^2(G(\mathbb{A})/G(\mathbb{Q})), \ell_X = P_X$. If X is involved with representation of G, to define P_X we need more data.

Example (the twisted (Fourier-Jacobi) GGP conjecture)

Let $H = \mathrm{U}(V) \to G = \mathrm{U}(V_K)$ ($K/\mathbb{Q}$ quadratic extension). Let ω be the Weil representation of H, for $\pi \in \mathcal{A}(G)$ we use

$$P_X(f \otimes \phi) = \int_{[H]} f(h)\overline{\phi(h)}dh \in \mathrm{Hom}_H(\pi, \omega),\ f \in \pi, \phi \in \omega$$

to detect Asai L-functions of π.

Producing arithmetic invariants from big G-representations

For arithmetic things, we can use

- **Representation from geometry**: use Shimura varieties Sh_G (arithmetic quotients of $G(\mathbb{R})/K_\infty$) over number fields with infinite levels. There is a natural action of $G(\mathbb{A}_f)$ on Sh_G hence on the cohomology ring $V_X^{(1)} := H^*(\mathrm{Sh}_G)$.
- **Arithmetic period integrals**: construct a natural arithmetic intersection pairing $(-,-) : H^*(\mathrm{Sh}_G) \times H^*(\mathrm{Sh}_G) \to \mathbb{C}$. **Construct X-special cycles** $\mathrm{Sh}_X \to \mathrm{Sh}_G$, and set $\ell_X^{(1)} = (-, \mathrm{Sh}_X)$.
- **Arithmetic Fundamental Lemma**: have a geometric decomposition of $\ell_X^{(1)}$ to intersections over local Shimura varieties, relate them to derivatives of orbital integrals.

Remarks on π_∞

We usually consider $\mathrm{U}(n-1,1)$-Shimura varieties, so $G_\infty = \mathrm{U}(n-1,1)$. To make π appears in $H^*(\mathrm{Sh}_G)$, we have restrictions on π_∞ (analogs of weight 2 cusp forms): via the base change maps

$$\mathrm{Rep}(\mathrm{U}(n-1,1)) \overset{BC}{\to} \mathrm{Rep}(\mathrm{GL}_n(\mathbb{C})) \overset{BC}{\leftarrow} \mathrm{Rep}(\mathrm{U}(n)).$$

we have $BC(\pi_\infty) = BC(1_{\mathrm{U}(n)})$. For higher weights, we have to use automorphic coefficients on Shimura varieties.
In general, if $\ell_X^{(1)}|_{\pi_f} \neq 0$ then $\mathrm{Hom}_{H(\mathbb{A}_f)}(\pi_f, 1) \neq 0$. By local GGP type conjecture, we expect

Conjecture

$\mathrm{Hom}_{H(\mathbb{R})}(\pi_\infty, 1) = 0$. And $L_{X^\vee}(\rho_\pi, 1/2) = 0$.

Hence it is natural to study higher derivatives $L'_{X^\vee}(\rho_\pi, 1/2)$.

Real mirabolic special cycles

Consider $G(\mathbb{R}) = \mathrm{GL}_n(\mathbb{R})$, choose a vector $u \in \mathbb{R}^n - 0$. We have the mirabolic subgroup

$$Mi(u) = \mathrm{Stab}(u) \leq G(\mathbb{R}).$$

Definition

The **mirabolic special cycle** for u is the image of $Mi(u)$ on the symmetric space $G(\mathbb{R})/K_\infty$, i.e. $Z(u) = \mathrm{Im}(Mi(u) \to G(\mathbb{R})/K_\infty)$.

Example: $n = 2$, get the line $y = 1$ on upper half plane $\mathbb{H}$. The (twisted) period integrals produce the Fourier coefficients of modular forms.

Real mirabolic special cycles

Consider $G(\mathbb{R}) = \mathrm{GL}_n(\mathbb{R})$, choose a vector $u \in \mathbb{R}^n - 0$. We have the mirabolic subgroup

$$Mi(u) = \mathrm{Stab}(u) \leq G(\mathbb{R}).$$

Definition

The **mirabolic special cycle** for u is the image of $Mi(u)$ on the symmetric space $G(\mathbb{R})/K_\infty$, i.e. $Z(u) = \mathrm{Im}(Mi(u) \to G(\mathbb{R})/K_\infty)$.

Example: $n = 2$, get the line $y = 1$ on upper half plane $\mathbb{H}$. The (twisted) period integrals produce the Fourier coefficients of modular forms.

Definition

Consider any real reductive group G, and an algebraic representation ρ of G on V. For $u \in V - 0$, the **mirabolic special cycle**

$$Z_V(u) \to G(\mathbb{R})/K_\infty$$

is the pullback of $Z(u)$ to $G(\mathbb{R})/K_\infty$ along $\rho : G \to \mathrm{GL}(V) \cong \mathrm{GL}_n(\mathbb{R})$.

p-adic local Shimura varieties

Let p be a prime, consider the unramified quadratic extension $\mathbb{Q}_{p^2}/\mathbb{Q}_p$.
Let V be a n-dimensional vector space over $\mathbb{Q}_{p^2}$.
We consider the local Shimura variety for $(\mathrm{GL}(V), \mu = (n-1, 1), b \,\text{basic})$

$$M_n \to \mathrm{Spf}\, \mathbb{Q}_{p^2}$$

which has a natural action of $J_b = \mathrm{GL}(\mathbb{V})$. Here $\mathbb{V}$ is another n-dimensional vector space over $\mathbb{Q}_{p^2}$.

p-adic local Shimura varieties

Let p be a prime, consider the unramified quadratic extension $\mathbb{Q}_{p^2}/\mathbb{Q}_p$.
Let V be a n-dimensional vector space over $\mathbb{Q}_{p^2}$.
We consider the local Shimura variety for $(\mathrm{GL}(V), \mu = (n-1,1), b \,\text{basic})$

$$M_n \to \mathrm{Spf}\, \mathbb{Q}_{p^2}$$

which has a natural action of $J_b = \mathrm{GL}(\mathbb{V})$. Here $\mathbb{V}$ is another n-dimensional vector space over $\mathbb{Q}_{p^2}$.
The rigid analytic variety M_n is smooth of dimension $2(n-1)$.
If there is no level, let

$$\mathcal{M}_n \to \mathrm{Spf}\, \mathbb{Z}_{p^2}$$

be its canonical smooth integral model which inherits the action of $\mathrm{GL}(\mathbb{V})$.

p-adic mirabolic special cycles

Proposition (Z.)

For $u \in \mathbb{V} - 0$, similarly there exists a Cartier divisor called mirabolic special cycle

$$Z(u) \to M_n.$$

We have

$$Z(g.u) = g.Z(u), \forall g \in \mathrm{GL}(\mathbb{V}).$$

If there is no level, the cycle descends to $\mathcal{Z}(u) \to \mathcal{M}_n$ with moduli interpretations.

Hence for general local Shimura datum (G, μ, b) with a map to $(\mathrm{GL}(V), \mu = (n-1, 1), b_0)$, we can pullback to get mirabolic special cycles on M_G for the representation V of G.

Unitary case: Kudla-Rapoport divisor

Assume moreover V is a split hermitian space over $\mathbb{Q}_{p^2}$. We consider the local Shimura variety for $\mathrm{U}(V), \mu = (n-1, 1), b \,\mathrm{basic})$

$$N_n \to \mathrm{Spf}\, \mathbb{Q}_{p^2}$$

which has a natural action of $\mathrm{U}(\mathbb{V})$. Here $\mathbb{V}$ is a n-dimensional non-split hermitian space over $\mathbb{Q}_{p^2}$. If there is no level, let

$$\mathcal{N}_n \to \mathrm{Spf}\, \mathbb{Z}_{p^2}$$

be its canonical smooth integral model which inherits the action of $\mathrm{U}(\mathbb{V})$. The rigid analytic variety N_n is smooth of dimension $n-1$ and are related to $\mathrm{U}(n-1, 1)$-Shimura varieties by uniformization. We have **a natural embedding**

$$\mathcal{N}_n \to \mathcal{M}_n$$

compatible with the action of $\mathrm{U}(\mathbb{V}) \to \mathrm{GL}(\mathbb{V})$.

Arithmetic intersections

Proposition

The divisor $\mathcal{Z}(u)|_{\mathcal{N}_n}$ is exactly the Kudla-Rapoport divisor studied in the Kudla program for arithmetic and geometric theta correspondences.

Note

$$\dim \mathcal{N}_n = (n-1)+1 = n, \dim \mathcal{Z}(u)|_{\mathcal{N}_n} = n-1.$$

As $n+(n-1) = 2n-1 = \dim \mathcal{M}_n$, for $(g,u) \in \mathrm{GL}(\mathbb{V}) \times \mathbb{V}$, we expect the intersection number

$$\mathrm{Int}(g\mathcal{N}_n, \mathcal{Z}(u)|_{\mathcal{N}_n})_{\mathcal{M}_n} \in \mathbb{Q} \cup \{\infty\}$$

is finite (or could be regularized), which is true if (g,u) is regular semi-simple.

Question: **is this local invariant automorphic**?

Twisted AFL

Theorem (Twisted AFL, Z.)

For any regular semisimple $(g,u)\in \mathrm{GL}(\mathbb{V})\times\mathbb{V}$*, we have*

$$\mathrm{Int}(g\mathcal{N}_n,\mathcal{Z}(u)|_{\mathcal{N}_n})_{\mathcal{M}_n}\log p=-\partial\mathrm{Orb}((\gamma,u'),F'_{\mathrm{std}}).$$

Here $(\gamma,u')\in \mathrm{GL}_n(\mathbb{Q}_p)\times\mathbb{Q}_p^n\times(\mathbb{Q}_p^n)^*$ *matches* (g,u)*.*

We consider the natural action of $h\in\mathrm{GL}_n(\mathbb{Q}_p)$ on

$$(\gamma,u')\in \mathrm{GL}_n(\mathbb{Q}_p)\times\mathbb{Q}_p^n\times(\mathbb{Q}_p^n)^*.$$

Set $F'_{\mathrm{std}}=1_{\mathrm{GL}_n(\mathbb{Z}_p)}\times\mathbb{Z}_p^n\times(\mathbb{Z}_p^n)^*$. Define orbital integrals ($s\in\mathbb{C}$)

$$\mathrm{Orb}(-,s)=\int_{h\in\mathrm{GL}_n(\mathbb{Q}_p)}F'_{\mathrm{std}}(h^{-1}\gamma h,h^{-1}.u')(-1)^{\mathrm{val}(\det h)}|\det h|^s dh.$$

Comparison between the p-adic and real world

- For $\mathrm{SL}_2(\mathbb{R})$, we see that $Z(u)$ on the upper half plane $\mathbb{H}$ or (modular curves) is a real line. After compactification we get a family of circles S^1 on compact Riemann surfaces, which is not algebraic (the complex dimension of S^1 shall be $1/2$). In general, real mirabolic special cycles are not algebraic (but the pullback of non-algebraic cycles could be algebraic).
- For $\mathrm{GL}_n(\mathbb{Q}_{p^2})$, we get algebraic p-adic mirabolic special cycles because of **the motivic doubling** (along the quadratic extension $\mathbb{Q}_{p^2}/\mathbb{Q}_p$ or $\mathbb{Q}_p \times \mathbb{Q}_p/\mathbb{Q}_p$). Idea: $S^1 \times S^1$ can be made a complex structure and algebraic. Hence we get codimension $1 = 1/2 + 1/2$ cycles on M_n. We could also define (non-algebraic) special cycles on any $G(\mathbb{Q}_p)/K_p$ or the Bruhat-Tits building of G.

Comparison between the p-adic and real world

- For $\mathrm{SL}_2(\mathbb{R})$, we see that $Z(u)$ on the upper half plane $\mathbb{H}$ or (modular curves) is a real line. After compactification we get a family of circles S^1 on compact Riemann surfaces, which is not algebraic (the complex dimension of S^1 shall be $1/2$). In general, real mirabolic special cycles are not algebraic (but the pullback of non-algebraic cycles could be algebraic).
- For $\mathrm{GL}_n(\mathbb{Q}_{p^2})$, we get algebraic p-adic mirabolic special cycles because of **the motivic doubling** (along the quadratic extension $\mathbb{Q}_{p^2}/\mathbb{Q}_p$ or $\mathbb{Q}_p \times \mathbb{Q}_p/\mathbb{Q}_p$). Idea: $S^1 \times S^1$ can be made a complex structure and algebraic. Hence we get codimension $1 = 1/2 + 1/2$ cycles on M_n. We could also define (non-algebraic) special cycles on any $G(\mathbb{Q}_p)/K_p$ or the Bruhat-Tits building of G.
- In automorphic representation theory, the analog is the Weil representation for $\mathrm{U}(V) \times \mathrm{U}(1)$ is not realizable on G-varieties, but the Weil representation for $\mathrm{U}(V) \times \mathrm{U}(1,1)$ could be realized on $X = V$. There is a doubling via partial Fourier transforms.

Thank you!